

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/200744645>

A Practical Guide to Spline

Book in Mathematics of Computation · January 1978

DOI: 10.2307/2006241

CITATIONS

4,805

READS

68,691

1 author:



Carl R de Boor

University of Wisconsin–Madison

247 PUBLICATIONS **25,004** CITATIONS

SEE PROFILE

Carl de Boor

A Practical Guide to Splines

Revised Edition

With 32 figures



Springer

Contents

Preface	v
---------	---

Notation	xv
----------	----

I · Polynomial Interpolation

Polynomial interpolation: Lagrange form	2
Polynomial Interpolation: Divided differences and Newton form	3
Divided difference table	8
Example: Osculatory interpolation to the logarithm	9
Evaluation of the Newton form	9
Example: Computing the derivatives of a polynomial in Newton form	11
Other polynomial forms and conditions	12
Problems	15

II · Limitations of Polynomial Approximation

Uniform spacing of data can have bad consequences	17
Chebyshev sites are good	20
Runge example with Chebyshev sites	22
Squareroot example	22
Interpolation at Chebyshev sites is nearly optimal	24
The distance from polynomials	24
Problems	27

III · Piecewise Linear Approximation

Broken line interpolation	31
Broken line interpolation is nearly optimal	32
Least-squares approximation by broken lines	32
Good meshes	35
Problems	37

IV · Piecewise Cubic Interpolation

Piecewise cubic Hermite interpolation	40
Runge example continued	41
Piecewise cubic Bessel interpolation	42
Akima's interpolation	42
Cubic spline interpolation	43
Boundary conditions	43
Problems	48

V · Best Approximation Properties of Complete Cubic Spline Interpolation and Its Error

Problems	56
----------	----

VI · Parabolic Spline Interpolation

Problems	64
----------	----

VII · A Representation for Piecewise Polynomial Functions

Piecewise polynomial functions	69
The subroutine PPVALU	72
The subroutine INTERV	74
Problems	77

VIII · The Spaces $\Pi_{<k,\xi,\nu}$ and the Truncated Power Basis

Example: The smoothing of a histogram by parabolic splines	79
The space $\Pi_{<k,\xi,\nu}$	82
The truncated power basis for $\Pi_{<k,\xi}$ and $\Pi_{<k,\xi,\nu}$	82
Example: The truncated power basis can be bad	85
Problems	86

IX · The Representation of PP Functions by B-Splines

Definition of a B-spline	87
Two special knot sequences	89
A recurrence relation for B-splines	89
Example: A sequence of parabolic B-splines	91
The spline space $S_{k,t}$	93
The polynomials in $S_{k,t}$	94
The pp functions in $S_{k,t}$	96
B stands for basis	99
Conversion from one form to the other	101
Example: Conversion to B-form	103
Problems	106

X · The Stable Evaluation of B-Splines and Splines

Stable evaluation of B-splines	109
The subroutine BSPLVB	109
Example: To plot B-splines	113
Example: To plot the polynomials that make up a B-spline	114
Differentiation	115
The subroutine BSPLPP	117
Example: Computing a B-spline once again	120
The subroutine BVALUE	121
Example: Computing a B-Spline one more time	126
Integration	127
Problems	128

XI · The B-Spline Series, Control Points, and Knot Insertion

Bounding spline values in terms of "nearby" coefficients	131
Control points and control polygon	133
Knot insertion	135
Variation diminution	138
Schoenberg's variation diminishing spline approximation	141
Problems	142

XII · Local Spline Approximation and the Distance from Splines

The distance of a continuous function from $S_{k,t}$	145
The distance of a smooth function from $S_{k,t}$	148
Example: Schoenberg's variation-diminishing spline approximation	149
Local schemes that provide best possible approximation order	152
Good knot placement	156

The subroutine NEWNOT	159
Example: A failure for NEWNOT	161
The distance from $\$_{k,n}$	163
Example: A failure for CUBSPL	165
Example: Knot placement works when used with a local scheme	167
Problems	169

XIII · Spline Interpolation

The Schoenberg-Whitney Theorem	171
Bandedness of the spline collocation matrix	173
Total positivity of the spline collocation matrix	169
The subroutine SPLINT	175
The interplay between knots and data sites	180
Even order interpolation at knots	182
Example: A large $\ I\ $ amplifies noise	183
Interpolation at knot averages	185
Example: Cubic spline interpolation at knot averages with good knots	186
Interpolation at the Chebyshev-Demko sites	189
Optimal interpolation	193
Example: "Optimal" interpolation need not be "good"	197
Osculatory spline interpolation	200
Problems	204

XIV · Smoothing and Least-Squares Approximation

The smoothing spline of Schoenberg and Reinsch	207
The subroutine SMOOTH and its subroutines	211
Example: The cubic smoothing spline	214
Least-squares approximation	220
Least-squares approximation from $\$_{k,t}$	223
The subroutine L2APPR (with BCHFAC/BCHSLV)	224
L2MAIN and its subroutines	228
The use of L2APPR	232
Example: Fewer sign changes in the error than perhaps expected	232
Example: The noise plateau in the error	235
Example: Once more the Titanium Heat data	237
Least-squares approximation by splines with variable knots	239
Example: Approximation to the Titanium Heat data from $\$_{4,9}$	239
Problems	240

XV · The Numerical Solution of an Ordinary Differential Equation by Collocation

Mathematical background	243
The almost block diagonal character of the system of collocation equations; EQBLOK, PUTIT	246
The subroutine BSPLVD	251
COLLOC and its subroutines	253
Example: A second order nonlinear two-point boundary-value problem with a boundary layer	258
Problems	261

XVI · Taut Splines, Periodic Splines, Cardinal Splines and the Approximation of Curves

Lack of data	263
“Extraneous” inflection points	264
Spline in tension	264
Example: Coping with a large endslope	265
A taut cubic spline	266
Example: Taut cubic spline interpolation to Titanium Heat data	275
Proper choice of parametrization	276
Example: Choice of parametrization is important	277
The approximation of a curve	279
Nonlinear splines	280
Periodic splines	282
Cardinal splines	283
Example: Conversion to ppform is cheaper when knots are uniform	284
Example: Cubic spline interpolation at uniformly spaced sites	284
Periodic splines on uniform meshes	285
Example: Periodic spline interpolation to uniformly spaced data and harmonic analysis	287
Problems	289

XVII · Surface Approximation by Tensor Products

An abstract linear interpolation scheme	291
Tensor product of two linear spaces of functions	293
Example: Evaluation of a tensor product spline.	297
The tensor product of two linear interpolation schemes	297
The calculation of a tensor product interpolant	299

Example: Tensor product spline interpolation	301
The ppform of a tensor product spline	305
The evaluation of a tensor product spline from its ppform	305
Conversion from B-form to ppform	307
Example: Tensor product spline interpolation (continued)	309
Limitations of tensor product approximation and alternatives	310
Problems	311

Postscript on Things Not Covered	313
----------------------------------	-----

Appendix: Fortran Programs

Fortran programs	315
List of Fortran programs	315
Listing of SOLVEBLOK Package	318

Bibliography	331
--------------	-----

Index	341
-------	-----